🛒 CAPSTONE PROJECT REPORT

Project 2: Online Retail Customer Segmentation

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a. Problem Statement

An online retail store is trying to understand the various purchase patterns of its customers to optimize marketing strategies and product offerings. E-commerce platforms deal with massive amounts of transaction data, and identifying individual customer behavior manually is impossible. As a Data Scientist, the goal is to provide evidence-based insights into customer purchasing history and perform customer segmentation.

b. Project Objective

- To clean and prepare large-scale transactional data (online_retail.csv containing **387,961 rows**) for analysis.
- To engineer an **RFM (Recency, Frequency, Monetary)** model to capture behavior profiles.
- To cluster and segment customers into distinct behavioral groups using the unsupervised **K-Means Clustering** machine learning algorithm.

c. Data Description

The dataset contains transaction history with 8 key attributes:

- **InvoiceNo:** A unique 6-digit identifier for each transaction.
- **StockCode:** A unique product/item ID.
- **Description:** The name of the product.
- **Quantity:** The number of units purchased per transaction.
- **InvoiceDate:** The exact date and time the invoice was generated.
- **UnitPrice:** The product price per single unit.
- **CustomerID:** A unique identifier assigned to each individual customer.
- **Country:** The name of the country where the customer resides.

d. Data Pre-processing Steps and Inspiration

1. **Handling Missing Values:** Inspected null records using df.isnull().sum(). Found major missing values in CustomerID (135,080 rows), which were safely dropped as customer metrics cannot be calculated without IDs.
2. **Filtering Anomalies:** Transactions with negative or zero values in Quantity and UnitPrice (which represent returns or errors) were removed to keep only valid purchases.
3. **Feature Engineering (RFM Metrics):**
    - **Recency:** Calculated by setting a reference date (latest_date = max date + 1 day) and measuring the days since the customer's last purchase.
    - **Frequency:** Calculated as the count of invoices generated per customer.

    - **Monetary:** Calculated by creating a total_amount column (Quantity * UnitPrice) and summing it up per customer.
4. **Outlier Treatment:** Applied extreme percentile filters (0.05 to 0.95) using the IQR/Tukey's method to eliminate canceled/bounced transactions and bulk wholesale outliers, preserving data integrity.
5. **Feature Scaling:** Since K-Means relies on Euclidean distance, StandardScaler was used to normalize the fields so that large monetary values do not distort the clustering.

e. Choosing the Algorithm for the Project

The **K-Means Clustering** algorithm was selected for this project. It is the gold standard for unsupervised market segmentation. The optimal number of clusters (K) was decided via the **Elbow Method (WCSS)**.

f. Motivation and Reasons For Choosing the Algorithm

- **Unsupervised Clustering:** The data does not have predefined target labels. K-Means group customers automatically based on mathematical similarities.
- **Scalability:** It is computationally fast and efficiently processes large RFM data arrays.
- **Interpretability:** The centroid averages can be directly translated into practical customer behavior segments (e.g., VIP, Churned, New).

g. Assumptions

- It is assumed that a customer's historical frequency and spending behavior are strong indicators of their future buying patterns.
- Active customers with a low recency value are assumed to have a higher probability of responding to retention campaigns.

h. Model Evaluation and Techniques

Using the Elbow Method, the distortion score flattened out, pointing to **K=5** as the ideal number of customer segments. The 5 clusters were profiled based on their RFM means:

- **Cluster 3: VIP / High-Value Customers** (Low Recency, High Frequency, High Monetary).
- **Cluster 0: High-Frequency Regulars** (Consistent buyers with moderate-to-high spending).
- **Cluster 4: Potential Loyalists** (Recent buyers with growing frequency).
- **Cluster 1: New / Low-Value Buyers** (Very recent, but have low spending and frequency so far).
- **Cluster 2: At-Risk / Lost Customers** (Very high Recency, haven't visited in months).

i. Inferences from the Same

- **Geographic Concentration:** The United Kingdom dominates total sales revenue, followed by the Netherlands and EIRE. Marketing focus should remain geo-targeted.
- **Seasonality Lift:** Sales velocity spikes massively during the Q4 period (September,

October, and November), signaling high year-end demand.
- **Peak Shopping Hours:** Transactions peak between 10:00 AM and 2:00 PM daily, providing the best window for sending real-time promotional push notifications.

j. Future Possibilities of the Project

- **Predictive Churn Modeling:** Using these cluster assignments as training labels to build a supervised classification model that predicts when a user is about to drift from Cluster 0 to Cluster 2.
- **Product Recommendation Integration:** Combining user cluster profiles with Market Basket Analysis to suggest tailored products during the checkout phase.
